JAN–I–8

29 APR. 1944

JOINT ARMY-NAVY SPECIFICATION

INSULATORS, STEATITE, RADIO

Army Number	Navy Number
71–4901	16–I–3

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and shall, unless otherwise directed, be used in lieu of the following specifications:

Signal Corps	Bureau of Ships
Tentative Specification 71–4901 1 Feb 1944	Ad Interim Specification 16–I–3 (INT) 1 Feb 1944

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–C–5—Capacitors, Mica-dielectric, Fixed.
JAN–I–10—Insulating Material, Ceramic Radio, Class L.

U. S. ARMY SPECIFICATIONS

(None)

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[1]
39P16—General Specification for Packaging and Packing for Overseas-Shipment.

B. TYPES.

B–1. Steatite radio insulators covered by this specification shall be furnished in the types shown on figures 5 to 13, inclusive, as specified in the contract or order. (See par. H–2a.)

[1] Applicable only to Navy purchases.

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C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—The material for each part shall be as hereinafter specified. (Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.)

C–1a. Steatite used in the construction of insulators shall be an approved grade conforming to the requirements of Joint Army-Navy Specification JAN–I–10. Where a definite grade of steatite is not specified, grade L–3 or better shall be used.

C–1b. Where a definite material for the hardware is not specified, a suitable material shall be used.

C–2. *Substitution of material.*—If the manufacturer desires to substitute for a specified or a previously approved material or fabricated part, he shall submit to the bureau or agency concerned with his bid or sample a statement to that effect, describing the proposed substitution, together with evidence to substantiate his claim that such substitute is suitable. Similar notification and substantiating evidence shall be submitted at any later time if substitution becomes necessary or desirable. At the discretion of the bureau or agency concerned, test samples may be required to prove the suitability of the proposed substitute.

C–3. *Screw threads.*—Where practicable, screw threads for all bolts, machine screws, nuts, and tapped holes shall be of National Form in accordance with the Screw-Thread Standards for Federal Services, Handbook H28, except as modified in table III for steatite. Unless otherwise specified, screws smaller than size 6-32 will not be permitted.

C–4. *Workmanship.*—Insulators shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and practice.

D. GENERAL REQUIREMENTS.

D–1. *Definitions.*—

D–1a. *Holes.*—

D–1a(1). *Mounting holes*—Mounting holes are defined as those holes in the insulator which are used as points of support for the insulator or for other members of an assembly.

D–1a(2). *Other holes.*—Other holes are defined as all holes other than mounting holes, such as winding or tap-off holes in coil forms, lead holes, perforations for ventilation or for weight reduction purposes, etc.

D–1b. *Blisters.*—Small bulges or lumps on the surface of a ceramic piece are known as blisters. The term does not necessarily indicate the presence of a cavity under the surface. (See par. D–2.)

D–1c. *Surface marks.*—Small discolored surface blemishes and color spots are known as surface marks. (See par. D.–2b.)

D–1d. *Camber.*—For the purposes of this specification, camber is defined in percent as the ratio $100\frac{e}{L}$ where e is the maximum perpendicular distance from the chord to the corresponding curved surface and L is a reference length measured between the sector extremities. (See figs. 1, 2, and 3 and par. E–5b(5).)

D–2. *Condition of surfaces.*—Surfaces which are chipped, blistered,

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or pimpled to the extent that the life and usefulness of the insulator are impaired shall be cause for rejection of the insulator. Rough or irregular edges caused by minute chips or nicks are not considered as impairing the life and usefulness of the insulator unless sharp edges are specified on the applicable figure. The following information shall be used in determining the significance of all surface imperfections:

D–2a. *Chips, blisters, pimples.*—Insulators having surfaces which are chipped, blistered, or pimpled shall be rejected in the event that such imperfections result in:

(1) Surfaces surrounding mounting holes which will not withstand the stress of mounting or will not provide secure mounting.

(2) Bearing surfaces which will not withstand bearing pressure or friction of associated assembly details.

(3) Surfaces of grooves or other indentations provided for carrying coil windings or other wiring which will not provide adequate support for winding or which will damage the wire.

(4) Sealing surfaces which will not provide sufficient support for materials used in sealing inserts or associated assembly details.

(5) Glazed surfaces which are not substantially free from chips occurring after glazing.

(6) Blisters which might easily become open cavities.

(7) Blisters which are excessively large or occur frequently on any surface so as to impair life and usefulness of the insulator.

NOTE.—It is deemed impractical to set up even general standards of quality for size and quantity of chips inasmuch as either must be judged in conjunction with the other. Since both size and application of steatite parts are subject to wide variation from one piece to another, it is believed that questions of size and quantity of chips should be considered on a common-sense basis rather than by any empirical formula covering "number of chips per square or linear inch" or "loss in volume."

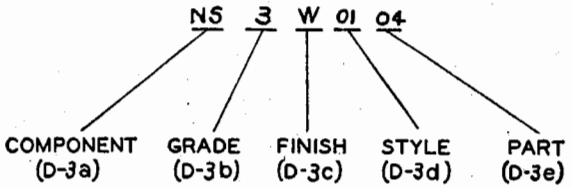
D–2b. *Surface marks.*—All insulators having surface marks shall be rejected in the event that the following conditions exist:

(1) Where there is evidence of a cavity.

(2) Where they occur frequently or are excessively large so as to impair life and usefulness of the insulator.

(3) Where they occur between coil-winding grooves and cover more than one-half the space between the grooves.

D–3. *Type designation.*—The type designation for steatite insulators is as follows:



D–3a. *Component.*—Steatite-body ceramic insulators are identified by the symbol "NS," where "N" indicates "insulator" and "S" indicates "steatite."

D–3b. *Grade.*—The grade of insulating material used is identified by a 1-digit number corresponding to the grades covered by Joint Army-Navy Specification JAN–I–10.

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D–3c. *Finish.*—The surface finish specified is identified by a single letter as follows:

Finish	*Designating letter*
Unglazed	U
White glaze	W
Brown glaze	B

NOTE.—Any special treatment or finish shall be specified in the contract or order. (See par. H–2d.)

D–3d. *Style.*—The style designation is a 2-digit number appearing on individual figures and identifying the general form of the insulator. Where a style number is less than 10, it shall be preceded by a zero to avoid confusion with the part number. (See par. D–3e.)

D–3e. *Part.*—The part number is a 2-digit number identifying individual insulators shown on individual figures. Where practicable, as in the case of stand-off and strain insulators, the part number will be based on the length of the part expressed in eighths of an inch.

E. DETAIL REQUIREMENTS.

E–1. Insulators shall be manufactured in accordance with the requirements of the applicable figure.

E–2. *Moisture-repellent treatment.*—Where a moisture-repellent treatment is specified, insulators shall be treated with a compound acceptable to the bureau or agency concerned.

E–3. *Fittings.*—Insulators shall be furnished with such fittings as may be specified on the applicable figure or in the contract or order. (See par. H–2b.) The methods employed for securing the fittings shall be of such nature as to insure that the completed insulator will withstand the specified tests.

E–4. *Tapped holes.*—All tapped holes, unless otherwise specified, shall be slightly countersunk.

E–5. *Tolerances.*—Unless otherwise specified, the following tolerances will be permitted on all dimensions of insulators:

E–5a. *Noncritical dimensions.*—All noncritical dimensions, which are indicated as such by common fractional dimensions on applicable figures or in the contract or order, may have tolerances of ± 2 percent of the dimension involved, but not more than $\pm \frac{1}{8}$ inch, on dimensions up to and including 12½ inches. On all dimensions above 12½ inches, a tolerance of ± 1 percent will be permitted. In no case need any tolerance be less than $\pm \frac{1}{64}$ inch.

E–5b. *Critical dimensions.*—All critical dimensions, which are indicated as such by decimal dimensions on applicable figures or in the contract or order, may have tolerances as follows:

E–5b(1). *Unglazed surfaces (general).*—A tolerance of ± 1 percent of the dimension involved, but not less than ± 0.005 inch, will be permitted between all unglazed surfaces. Unless otherwise specified, a minimum tolerance of ± 0.010 inch will be permitted on thickness dimensions.

E–5b(2). *Glazed surfaces (general).*—A tolerance of ± 2 percent of the dimension involved, but not less than ± 0.012 inch, will be permitted between all glazed surfaces. This tolerance shall apply to any dimension referenced to a glazed surface.

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E–5b(3). *Cylindrical shapes—outside and inside diameters (unglazed).*—A tolerance of ±1 percent of the dimension involved, but not less than ±0.005 inch, will be permitted on all unglazed diameters. Unless otherwise specified, a minimum tolerance of ±0.010 inch will be permitted on nominal wall thickness dimensions.

E–5b(4). *Cylindrical shapes—outside and inside diameters (glazed).*—A tolerance of ±2 percent of the dimension involved, but not less than ±0.012 inch, will be permitted on all diameters up to and including 1 inch. Where the diameter is greater than 1 inch, (±1 percent of the diameter involved +0.010 inch) will be permitted. Unless otherwise specified, a minimum tolerance of ±0.010 inch will be permitted on all nominal wall thickness dimensions.

Example:

Glazed tubes (inches)	*Tolerances (± inch)*
O. D. 0.500	O. D. 0.012
I. D. .125	I. D. .012
O. D. 1.125	O. D. .021
I. D. .825	I. D. .0165
O. D. 2.00	O. D. .030
I. D. 1.750	I. D. .0275

E–5b(5). *Camber.*—The camber (see par. D–1d) shown in table I shall apply to all surfaces of all shapes. Figures 1, 2, and 3 shall be used to establish the ratios listed in table I. For flat shapes, disks, plates, and the like, these limits shall apply only where the thickness is equal to or greater than the thickness determined by the formulas given in paragraph 1b(10) of the appendix. Where the thickness is less than that determined from paragraph 1b(10) of the appendix, the camber shall be specified on the applicable figure or in the contract or order. (See par. H–2e.)

TABLE I

$\frac{L}{a}$	Maximum camber (percent of L) to be applied to surfaces	
	Unglazed	*Glazed*
Less than 2	0.25	0.35
From 2 to less than 3	.30	.40
From 3 to less than 5	.35	.45
From 5 to less than 8	.40	.50
From 8 to less than 13	.45	.55
From 13 to less than 20	.50	.60
From 20 to less than 30	.55	.65
From 30 to less than 40	.60	.70
From 40 to less than 50	.65	.75

NOTE.—In determining "maximum camber" permitted, values of e (figs. 1, 2, and 3) less than 0.006 inch shall be disregarded, unless otherwise specified.

E–5b(6). *Ellipticity of tubes.*—The ellipticity shall be determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient shall not exceed 1.02 and shall apply to cylindrical shapes having a wall thickness

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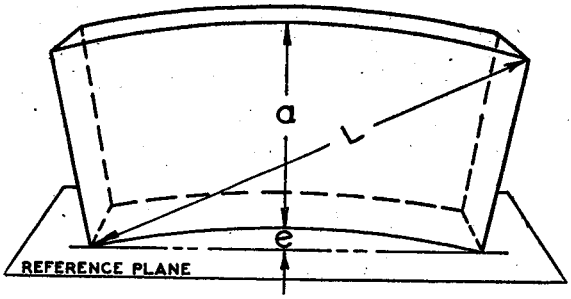


FIGURE 1.

L=Maximum diagonal length on the face.
a=Minimum dimension in same direction as *e*.

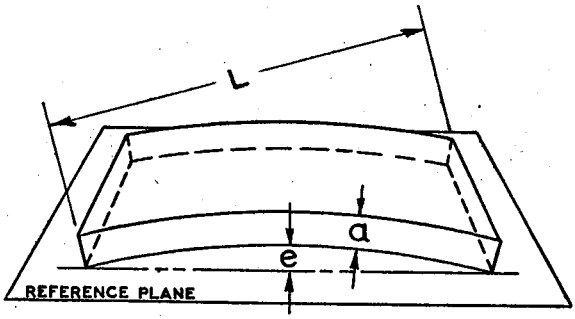


FIGURE 2.

L=Maximum diagonal chord length.
a=Minimum dimension in same direction as *e*.

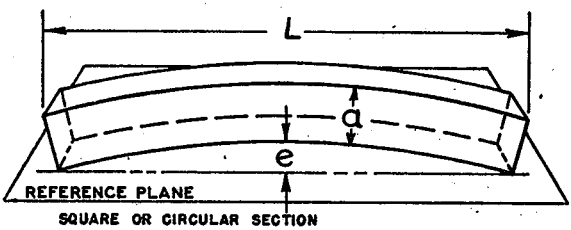


FIGURE 3. Square or circular section.

L=Maximum length of piece.
a=Minimum diameter or minimum dimension in same direction as *e*.

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of at least 10 percent of the outside diameter. If the wall thickness at the thinnest section of the wall is less than 10 percent of the outside diameter, the same ratio shall not exceed 1.03.

E–5b(7). *Hole centers.*—Standard tolerance of ± 0.005 inch for the dimensions between center of holes for all hole-center spacings less than 0.500 inch, and ± 1 percent of the dimension involved for all hole-center spacings of 0.500 inch and greater, will be permitted.

NOTE.—If pin gages are used for inspecting hole-center spacing, the design of the gage should be such as to meet the specific requirements. In using pin gages, consideration should be given to the tolerance between hole-centers as well as hole-diameter tolerance.

E–5b(8). *Hole diameters (unglazed).*—Standard tolerance of ± 0.005 inch for all hole diameters up to 0.500 inch, and ± 1 percent of the diameter for all hole diameters 0.500 inch and greater will be permitted. Holes which are not perfect circles shall have the same tolerance applied to the minor axis only.

NOTE.—Holes leading from glazed, surfaces shall have the tolerances for glazed surfaces applied except when specified that glaze be removed from the hole, in which case the tolerances for unglazed surfaces shall apply.

E–5b(9). *Hole diameters (glazed).*—Standard tolerance of ± 0.012 inch for all hole diameters up to 0.600 inch, and ± 2 percent of the diameter for all hole diameters 0.600 inch and greater will be permitted. Holes which are not perfect circles shall have the same tolerance applied to the minor axis only.

E–5b(10). *Parallel limits.*—Parallelism of surfaces will be considered satisfactory if the thickness measured at any point is within the established thickness tolerance of the piece.

E–5b(11). *Pillar insulators.*—Parallelism of top and bottom surfaces of pillar insulators shall be within 2 degrees, and these surfaces shall be perpendicular to the center axis within 2 degrees.

E–5c. *Tapped holes.*—When checked as described in paragraph F–5b, tapped holes shall be within the limits shown in table III.

E–6. *Thermal shock.*—When specified on the applicable figure or in the contract or order (see par. H–2c), insulators shall be capable of withstanding the thermal shock test described in paragraph F–6. The number of failures due to mechanical damage shall not exceed those permitted in table II, nor shall the glazed surfaces of samples subjected to the thermal shock test show evidence of crazing.

E–7. *Mechanical load.*—When specified on the applicable figure or in the contract or order (see par. H–2c) insulators shall be capable of withstanding the mechanical load specified in paragraph F–7.

E–8. *High potential.*—When specified on the applicable figure or in the contract or order (see par. H–2c), insulators shall be capable of withstanding the high-potential test voltages specified in paragraph F–8.

E–9. *Galvanizing (zinc coating).*—Galvanizing on metal parts shall meet the Preece test requirements.

E–10. *Other requirements.*—If additional requirements are given on the applicable figure or in the contract or order (see par. H–2c), insulators shall be capable of complying with the specified tests (see par. F–10).

E–11. *Marking.*—

E–11a. *Unglazed insulators.*—Each insulator shall be permanently marked with proper type designation (see par. D–3) and the manu-

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facturer's name or symbol. Such markings shall be accomplished by means of stamping or molding all necessary markings on the body of the insulator.

E–11b. *Glazed insulators.*—Each insulator shall be permanently marked with proper type designation (see par. D–3) and the manufacturer's name or symbol. Such markings shall be accomplished by means of stamping in or below the glaze, or by molding or etching when and where specifically approved. All necessary markings shall be applied on the body of the insulator.

E–11c. *Marking of cross sections.*—Items such as coil tubes, which are standardized only with respect to cross section and which are not required to carry individual part numbers, may be marked with the component and grade symbols of the type designation. (See par. D–3.)

Example: NS 3.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Test equipment and inspection facilities.*—The insulator manufacturer shall furnish and maintain all necessary facilities and equipment for making all tests and inspection which are made at the place of manufacture and shall perform all tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity to enable the inspection to keep up with production. The test equipment shall be of sufficient accuracy and quality to permit performance of the required tests. Where possible, the inspection of insulators shall be at the place of manufacture.

F–2. *Special gages.*—If special gages for inspection of parts are to be used by the bureau or agency concerned, requirements for such gages or a set of gages themselves, if practicable, should be furnished to the manufacturer by the bureau or agency concerned.

F–3. *Production tests.*—Each insulator shall be subjected to visual inspection and to other tests, as specified on the applicable figure or in the contract or order (see par. H–2c). Failures on production tests shall result in the rejection of the item failing, unless it can be shown to the satisfaction of the Government inspector that the items can be corrected with respect to the substandard features, in which case the items shall be corrected and reinspected.

F–4. *Production-sampling tests.*—For production sampling, insulators shall be selected by the Government inspector from insulators that have passed production tests. Table II lists the quantities of insulators to be selected. All tests, except visual inspection, shall be made only when specified on the applicable drawing furnished with the contract or order. Insulators shall be subjected to the applicable tests, in the order shown, and there shall be no more failures than the number specified. If the number of failures exceeds the maximum number permitted in table II, a retest shall be made. If the number of failures of the same type still exceeds the maximum allowable, this shall not necessarily be cause for complete rejection. If the failures on the retest exceed 2 percent of the entire lot inspected, the entire lot shall be subjected to individual tests, and the items failing shall be rejected. If the number of failures does not exceed 2 percent, the lot shall be accepted, unless, in the opinion of the Government inspector, the nature of the defect is detrimental to the life or performance of the part, in which case the entire lot shall be subjected to individual inspection.

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TABLE II

Test	Test requirements	Test procedure	Quantity in continuous production									
			0-50		51-100		101-500		501-5,000		Each additional 5,000	
			A	B	A	B	A	B	A	B	A	B
Visual inspection	D-3, E-1, E-3, E-4, E-5, E-11	F-5	0	0	4	0	6	0	12	0	12	0
Thermal shock	E-6	F-6	0	0	4	1	6	1	12	2	12	2
Mechanical load	E-7	F-7	As specified									
High potential	E-8	F-8	As specified									
Galvanizing	E-9	F-9	0	0	4	0	6	0	12	0	12	0
Other	E-10	F-10	As specified									

Where:
A = Number of samples required for test.
B = Maximum number of failures permitted.

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F–5. *Visual inspection.*—Insulators shall be inspected to determine that the workmanship is satisfactory (see pars. D–3, E–5, and E–11), and that the quantity is correct.

F–5a. *Tolerances.*—Insulators shall be inspected to determine that their physical dimensions are within the specified limits. (Where specific tolerances are not given on the applicable figure, the general tolerances specified in par. E–5 shall apply.)

F–5b. *Inspection of tapped holes.*—The following gages shall be used for the inspection of threads in tapped holes:

(1) For checking the minor diameter, d (see fig. 4): A plain Go gage pin, diameter A, and a plain Not Go gage pin, diameter B.

(2) A Go thread plug gage.

NOTE.—See table III for gage dimensions.

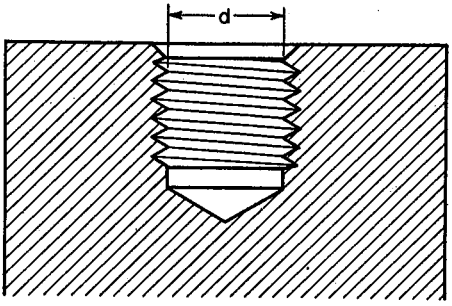


FIGURE 4.—Inspection of tapped holes.

TABLE III

Size of tapped hole	Column 1	Column 2	Column 3	Threads per inch	Acceptable gaging combinations
	Go gage pin, diameter A	Not Go gage pin, diameter B	Go thread plug gage		
	Inch	*Inch*	*Inch*		
4–40 [1]	0.082	0.098	0.112	40	Columns 1, 2, and 3 or columns 2 and 3 only.
6–32	.104	.122	.138	32	
8–32	.130	.148	.164	32	
10–32	.157	.174	.190	32	
¼–20	.195	.222	.250	20	
5/16–18	.252	.281	.312	18	
⅜–16	.307	.337	.375	16	

[1] Where specific approval has been obtained for the use of 4–40 tapped holes, these limits shall apply.

F–5c. *Inspection of external screw threads.*—Unless otherwise specified, the manufacturing limits shall be those for class 1 screws, coarse-thread and fine-thread series (Handbook H28) with the following exceptions:

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(1) The pitch diameter, lead, and thread angle will be acceptable if the finished steatite screw enters for its full length a gage representing a class 1 minimum nut (Handbook H28).

(2) The maximum minor diameter of steatite screws shall be 0.005 inch less than the minimum minor diameter of a class 1 nut (Handbook H28).

(3) The fit of steatite screws in a class 1 maximum nut shall give a thread depth engagement at least equal to the minimum thread depth of a class 1 fit between a screw and nut (National Bureau of Standards Handbook H28). This is determined as follows:

$$\frac{(a-b)100}{d} = \text{percent thread depth engagement}$$

Where:

a = Outside diameter of steatite thread.
b = Maximum minor diameter of a class 1 nut.
d = Double depth of the thread.

F–6. *Thermal shock test.*—

F–6a. Sample insulators shall be immersed in an ice-water bath for 10 minutes and then transferred as quickly as possible to a hot-water bath and allowed to remain there for 10 minutes. The samples shall be transferred back to the ice-water bath, and the thermal cycles shall be continued until each sample breaks or has successfully withstood the required number of cycles. Unless otherwise specified, the test specimens shall be exposed to 20 thermal cycles. The hot-water bath shall be maintained at not lower than 94° C. (201° F.) and the cold-water bath at not higher than 4° C. (39° F.).

F–6b. *Glazed insulators.*—After sample insulators have been subjected to the required number of thermal cycles, they shall be dried; then the surface shall be wetted completely by a blue-black ink, either by immersion or by brushing. The surface shall then be wiped dry with a clean cloth, and the specimens examined for crazing of the glaze under a low-power microscope (magnification of 20 diameters).

F–7. *Mechanical load test.*—Insulators shall be tested to determine their ability to withstand the type and magnitude of loading as specified on the applicable figure or in the contract or order. (See par. H–2c.)

F–8. *High-potential test.*—Insulators shall be subjected to a high-potential test to determine their ability to withstand without breakdown the voltage specified on the applicable figure or in the contract or order. (See par. H–2c.)

F–9. *Galvanizing (Zinc coating).*—Insulators shall be tested for adequate galvanizing by subjecting them to the Preece test.

F–10. *Other tests.*—Insulators shall be subjected to such additional tests as may be required on the applicable figure or in the contract or order. (See par. H–2c.)

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *U. S. Army.*—Shipments of components shall be packaged, packed, and marked as specified in the bid, request, contract, or basic equipment specification.

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G–2. *Navy.*—
G–2a. *Packaging.*—
G–2a(1). *Bulk shipment.*—
G–2a(1)a. Unless otherwise specified, insulators, except metal base cylindrical pillar insulators (fig. 8) and tubes, coil and miscellaneous insulators (fig. 12), shall be bulk packaged in bags fabricated of cloth, paper, or a combination of both. A uniform count (not to exceed 100 per bag) shall be maintained.
G–2a(1)b. Unless otherwise specified, metal base cylindrical pillar insulators (fig. 8) and tubes, coil, and miscellaneous insulators (fig. 12) shall require no packaging.
G–2a(2). *Individual shipment.*—Unless otherwise specified, the subject commodity, when shipped as replacement parts, individual units, or as kit groups, shall be packaged individually in a reverse tuck, straight tuck, or two-piece telescope style folding carton. The minimum caliper of board shall be 0.020-inch. Each box shall contain one insulator and shall be designed to fit the insulator in a compact manner.
G–2b. *Packing.*—
G–2b(1). *For domestic shipment.*—
G–2b(1)a. *Bulk shipment.*—
G–2b(1)a(1). Unless otherwise specified, the subject commodity, packaged as described in paragraph G–2a(1)a shall be packed in style R. S. C. 200-pound Mullen test corrugated or solid fiberboard boxes. Bags shall be cushioned within the shipping container by means of suitable cushioning material. The gross weight shall not exceed 40 pounds.
G–2b(1)a(2). Unless otherwise specified, metal base cylindrical pillar insulators (fig. 8) and tubes, coil and miscellaneous insulators (fig. 12) shall be packed in a style R. S. C. 200-pound Mullen test corrugated or solid fiberboard box employing cell partitions. Each cell shall contain one insulator. When two or more tiers of partitions are used in a container, dividing or separating pads of corrugated fiberboard shall be placed between each tier. All flaps shall be securely sealed in such a manner as to permit opening for inspection and subsequent reclosure without damage to the flaps. The gross weight shall not exceed 40 pounds.
G–2b(1)b. *Individual shipment.*— Unless otherwise specified, the subject commodity, packaged as described in paragraph G–2a(2), shall be overpacked in a style R. S. C. 200-pound Mullen test corrugated or solid fiberboard box. In order to permit inspection of the contents, the boxes shall be securely sealed in such a manner as to permit opening and subsequent reclosure without damage to the upper flaps. The gross weight shall not exceed 40 pounds.
G–2b(2). *For overseas shipment.*— Unless otherwise specified, the subject commodity, packed as described in paragraph G–2b(1)b, shall be overpacked in cleated plywood boxes, nailed wood boxes, or weatherproof fiberboard boxes conforming to the requirements of sections V, VI, and VIII, respectively, of Navy Department Specification 39P16. Each box shall be lined with a sealed waterproof bag, or its equivalent, made in conformance with the construction and tests of section XXV. The material used shall have a minimum water resistance of 12 hours. Each box shall be strapped in accordance with the requirements of the applicable section of Navy Department Specification 39P16. The gross weight shall not exceed approximately 75 pounds.

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G–2c. *Marking.*—

G–2c(1). *Packages.*— Unless otherwise specified, bulk carrying bags shall be conspicuously marked or tagged in one place, and individual folding cartons conspicuously marked on each of two adjacent sides in black with the following information completed:

Article_____
Type No_____
Stock No_____
Order No_____
Inspection_____
Contractor_____
Manufacturer_____
Quantity_____

G–2c(2). *Shipping containers.*—Unless otherwise specified, shipping containers shall be conspicuously marked in waterproof black on two adjacent sides, with the following information completed, as indicated:

Article_____
Type No_____
Stock No_____
Order No_____
Quantity_____
Contractor_____
Manufacturer_____
Gross Weight_____
Fragile—Handle with care_____

In addition to the foregoing, marking shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders must contain the title of the specification, the number, and date.

H-2. Requests, requisitions, schedules, and contracts or orders should cover the following features:

(*a*) Type of insulator desired.
(*b*) Desired fittings not shown on applicable figures.
(*c*) Any additional requirements or tests desired.
(*d*) Any special treatment or finish desired.
(*e*) Any special camber for shapes.
(*f*) Packaging, packing, and marking requirements for Army purchases. (*See* par. G–1.)

H–3. This specification is not intended to be restrictive with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered provided the bid clearly describes the specific points in which the construction proposed differs from that described herein, and provided further, that all performance tests and other requirements of this specification are met. When exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering insulators in strict accordance with the requirements of this specification.

H–4. This specification has been prepared through the coordinated effort of representatives of industry and the War and Navy Depart-

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ments. This specification should be used in the design of new equipment and that insofar as is practicable, replacement parts should also conform to these requirements.

H–5. Many additional sizes and shapes of steatite radio insulators not covered by this specification are now in use. Existing designs under which equipment is being, and will continue to be fabricated, are such as not to permit complete substitution with insulators conforming to the requirements of this specification, neither are these requirements suitable in all cases for direct replacement of existing insulators.

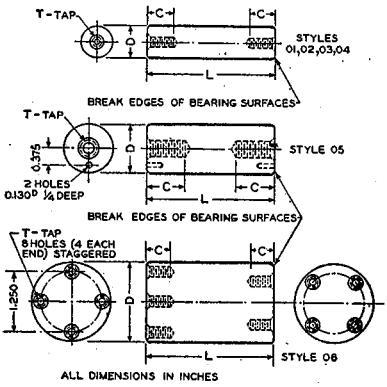
H–6. This specification is identical in all essential technical respects to the October 6, 1943, issue of the American War Standards C75.2—1943.

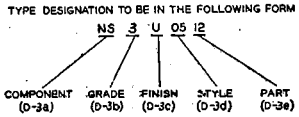
(*a*) Reference to American War Standards C75.1—1943 and C75.3—1942 have been deleted and Joint Army-Navy Specifications JAN–I–10 and JAN–C–5, which are identical in all essential technical respects, are referenced.

(*b*) Paragraphs on approval of material have been deleted as approval is covered in the material specification.

H–7. Matters relating to the use of components furnished in accordance with the requirements of this specification for use in connection with the design, production, and procurement of all Naval Radio, Radar, and Underwater Sound Equipment on the Electronics Precedence List should be referred to the Bureau of Ships Radio Division, Navy Department, Washington 25, D. C.

H–8. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.





Heavy lines indicate glaze when specified.
Countersink all tapped holes.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

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Type Designation†	Dimensions (inches)				Minimum usable threads	Body of insulator of grade§		
						L-3	L-4	L-5
	L	C	D	T		Available in finishes of—		
NS--0104	0.500	5/32	3/8	6-32	4	UW	UW	UW
NS--0105	0.625	5/16	3/8	6-32	5	UW	UW	UW
NS--0106	0.750	1/4	3/8	6-32	6	UW	UW	UW
NS--0108	1.000	3/8	3/8	6-32	6	UW	UW	UW
NS--0110	1.250	3/8	3/8	6-32	6	UW	UW	UW
NS--0112	1.500	3/8	3/8	6-32	6	UW	UW	UW
NS--0116	2.000	3/8	3/8	6-32	6	UW	UW	UW
NS--0205	0.625	9/16	1/2	8-32	5	UW	UW	UW
NS--0206	0.750	1/4	1/2	8-32	6	UW	UW	UW
NS--0208	1.000	3/8	1/2	8-32	6	UW	UW	UW
NS--0210	1.250	3/8	1/2	8-32	6	UW	UW	UW
NS--0212	1.500	3/8	1/2	8-32	6	UW	UW	UW
NS--0216	2.000	3/8	1/2	8-32	6	UW	UW	UW
NS--0220	2.500	3/8	1/2	8-32	6	UW	UW	UW
NS--0224	3.000	3/8	1/2	8-32	6	UW	UW	UW
NS--0308	1.000	3/8	3/4	10-32	6	UW	UW	UW
NS--0310	1.250	3/8	3/4	10-32	6	UW	UW	UW
NS--0312	1.500	3/8	3/4	10-32	6	UW	UW	UW
NS--0316	2.000	3/8	3/4	10-32	6	UW	UW	UW
NS--0320	2.500	3/8	3/4	10-32	6	UW	UW	UW
NS--0324	3.000	3/8	3/4	10-32	6	UW	UW	UW
NS--0332	4.000	3/8	3/4	10-32	6	UW	UW	UW
NS--0410	1.250	7/16	1	1/4-20	6	UW	UW	UW
NS--0412	1.500	1/2	1	1/4-20	6	UW	UW	UW
NS--0416	2.000	5/8	1	1/4-20	6	UW	UW	UW
NS--0420	2.500	5/8	1	1/4-20	6	UW	UW	UW
NS--0424	3.000	5/8	1	1/4-20	6	UW	UW	UW
NS--0432	4.000	5/8	1	1/4-20	6	UW	UW	UW
NS--0440	5.000	5/8	1	1/4-20	6	UW	UW	UW
NS--0448	6.000	5/8	1	1/4-20	6	UW	UW	UW
NS--0512	1.500	1/2	1 1/4	1/4-20	6	UW	UW	UW
NS--0516	2.000	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0520	2.500	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0524	3.000	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0532	4.000	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0540	5.000	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0548	6.000	5/8	1 1/4	1/4-20	6	UW	UW	UW
NS--0608	1.000	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0610	1.250	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0612	1.500	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0616	2.000	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0620	2.500	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0624	3.000	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0632	4.000	3/8	1 3/4	10-32	6	UW	UW	UW
NS--0640	5.000	3/8	1 3/4	10-32	6	UW	UW	UW

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-). The basic style number is given in the sketch. Insulators of the same basic design but with larger cross-sectional areas are indicated by the larger (succeeding) style numbers given in the tabulation.

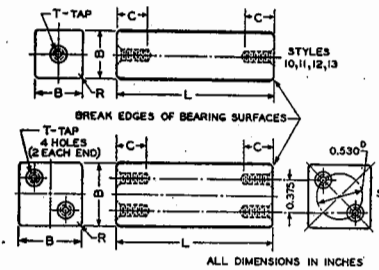
§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

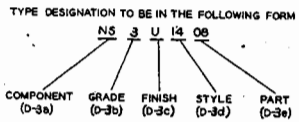
FIGURE 5.—Steatite radio insulators—cylindrical pillar insulators.

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H–9. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of United States Army and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy and United States Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–10. Copies of Joint Army-Navy specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.





Heavy lines indicate glaze when specified.
Countersink all tapped holes.
All edges perpendicular to bearing surfaces shall have the following nominal radii:

Dimension B (See below)	*Radius—R*
⅜	Max—1/32
½	Max—1/16
¾	Max—⅛, Min—1/16
1	Max—⅛, Min—1/16

Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

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Type designation†	Dimensions (inches)				Minimum usable threads	Body of insulator of grade§		
						L-3	L-4	L-5
	L	B	C	T		Available in finishes of—		
NS--1004	0.500	3/8	5/32	6-32	4	UW	UW	UW
NS--1005	0.625	3/8	3/16	6-32	5	UW	UW	UW
NS--1006	0.750	3/8	1/4	6-32	6	UW	UW	UW
NS--1008	1.000	3/8	3/8	6-32	6	UW	UW	UW
NS--1010	1.250	3/8	3/8	6-32	6	UW	UW	UW
NS--1012	1.500	3/8	3/8	6-32	6	UW	UW	UW
NS--1016	2.000	3/8	3/8	6-32	6	UW	UW	UW
NS--1106	0.750	1/2	1/4	8-32	6	UW	UW	UW
NS--1108	1.000	1/2	3/8	8-32	6	UW	UW	UW
NS--1110	1.250	1/2	3/8	8-32	6	UW	UW	UW
NS--1112	1.500	1/2	3/8	8-32	6	UW	UW	UW
NS--1116	2.000	1/2	3/8	8-32	6	UW	UW	UW
NS--1120	2.500	1/2	3/8	8-32	6	UW	UW	UW
NS--1208	1.000	3/4	3/8	10 32	6	UW	UW	UW
NS--1210	1.250	3/4	3/8	10-32	6	UW	UW	UW
NS--1212	1.500	3/4	3/8	10-32	6	UW	UW	UW
NS--1216	2.000	3/4	3/8	10-32	6	UW	UW	UW
NS--1220	2.500	3/4	3/8	10-32	6	UW	UW	UW
NS--1224	3.000	3/4	3/8	10-32	6	UW	UW	UW
NS--1232	4.000	3/4	3/8	10-32	6	UW	UW	UW
NS--1312	1.500	1	1/2	1/4-20	6	UW	UW	UW
NS--1316	2.000	1	5/8	1/4-20	6	UW	UW	UW
NS--1320	2.500	1	5/8	1/4-20	6	UW	UW	UW
NS--1324	3.000	1	5/8	1/4-20	6	UW	UW	UW
NS--1332	4.000	1	5/8	1/4-20	6	UW	UW	UW
NS--1408	1.000	1	3/8	8-32	6	UW	UW	UW
NS--1410	1.250	1	3/8	8-32	6	UW	UW	UW
NS--1412	1.500	1	3/8	8-32	6	UW	UW	UW
NS--1416	2.000	1	3/8	8-32	6	UW	UW	UW
NS--1420	2.500	1	3/8	8-32	6	UW	UW	UW
NS--1424	3.000	1	3/8	8-32	6	UW	UW	UW
NS--1432	4.000	1	3/8	8-32	6	UW	UW	UW
NS--1440	5.000	1	3/8	8-32	6	UW	UW	UW
NS--1448	6.000	1	3/8	8-32	6	UW	UW	UW

† Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-). The basic style is given in the sketch. Insulators of the same basic design but with larger cross-sectional areas are indicated by the larger (succeeding) style numbers given in the tabulation.

§ In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

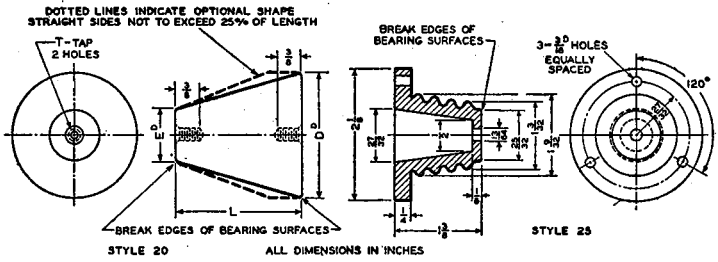
FIGURE 6.—Steatite radio insulators—square pillar insulators.

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TYPE DESIGNATION TO BE IN THE FOLLOWING FORM

NS 3 U 20 08

COMPONENT (D-3a) | GRADE (D-3b) | FINISH (D-3c) | STYLE (D-3d) | PART (D-3e)



Heavy lines indicate glaze when specified.
Glaze on inside of style 25 insulator optional with the manufacturer.
Countersink all tapped holes.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

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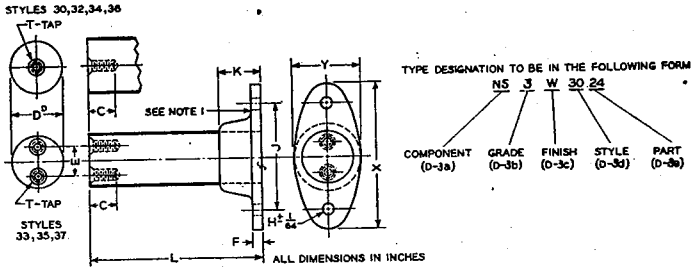
Type designation†	Dimensions (inches)				Minimum usable threads	Body of insulator of grade§		
						L–3	L–4	L–5
	L	D	E	T		Available in finishes of—		
NS- -2008	1.000	1	½	8–32	6	UW	UW	UW
NS- -2012	1.500	1	½	8–32	6	UW	UW	UW
NS- -2016	2.000	1⅛	⅝	10–32	6	UW	UW	UW
NS- -2024	3.000	1½	¾	10–32	6	UW	UW	UW
NS- -2501	----------	--------	--------	--------	----------	BW	BW	BW

† Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-).

§ In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

FIGURE 7.—Steatite radio insulators—conical pillar insulators.

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Base shall be of malleable iron or steel (hot galvanized), and shall be permanently attached to the insulator.
Bottom of base to be finished before galvanizing.
Samples shall be tested to determine compliance with E-9.
Top of insulator and bottom of base shall be parallel to each other and perpendicular center axis.
Heavy lines indicate glaze.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.
NOTE 1.—For bolt-head clearance at mounting holes (Column H), the minimum radius of flat area measured from center of hole shall be equal to the value of "H" listed below.

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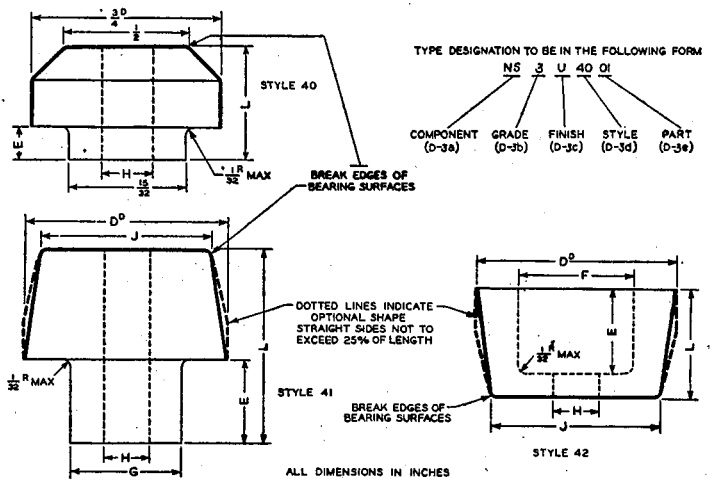
Type designation†	Dimensions (inches)											Number of tapped holes	Minimum usable threads	Body of insulator of grade §		
	L±1/64	D	C	T	E	F max.	H ±1/64	K max.	J	X max.	Y max.			L-3	L-4	L-5
														Available in finishes of—		
NS- -3024	3.000	1	5/8	1/4-20	------	5/16	9/32	1	2 1/4	3 1/4	1 3/4	1	6	W	W	W
NS- -3032	4.000	1	5/8	1/4-20	------	5/16	9/32	1	2 1/4	3 1/4	1 3/4	1	6	W	W	W
NS- -3040	5.000	1	5/8	1/4-20	------	5/16	9/32	1	2 1/4	3 1/4	1 3/4	1	6	W	W	W
NS- -3232	4.000	1 1/4	5/8	1/4-20	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3240	5.000	1 1/4	5/8	1/4-20	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3248	6.000	1 1/4	5/8	1/4-20	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3332	4.000	1 1/4	5/8	10-32	13/16	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3340	5.000	1 1/4	5/8	10-32	13/16	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3348	6.000	1 1/4	5/8	10-32	13/16	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3440	5.000	1 1/2	3/4	3/8-16	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3448	6.000	1 1/2	3/4	3/8-16	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3464	8.000	1 1/2	3/4	3/8-16	------	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	1	6	W	W	W
NS- -3540	5.000	1 1/2	5/8	10-32	1	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3548	6.000	1 1/2	5/8	10-32	1	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3564	8.000	1 1/2	5/8	10-32	1	5/16	11/32	1 1/4	2 5/8	3 17/32	2 1/16	2	6	W	W	W
NS- -3640	5.000	1 3/4	3/4	3/8-16	------	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	1	6	W	W	W
NS- -3648	6.000	1 3/4	3/4	3/8-16	------	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	1	6	W	W	W
NS- -3664	8.000	1 3/4	3/4	3/8-16	------	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	1	6	W	W	W
NS- -3740	5.000	1 3/4	3/8	10-32	1 1/4	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	2	6	W	W	W
NS- -3748	6.000	1 3/4	3/8	10-32	1 1/4	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	2	6	W	W	W
NS- -3764	8.000	1 3/4	3/8	10-32	1 1/4	5/16	13/32	1 9/16	3 3/4	5 1/8	2 13/16	2	6	W	W	W

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-). The basic style number is given in the sketch. Insulators of the same basic design but with larger cross-sectional areas are indicated by the larger style numbers given in the tabulation.

§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

FIGURE 8.—Steatite radio insulators—metal-base, cylindrical pillar insulators.

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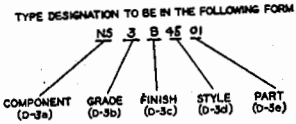
Heavy lines indicate glaze when specified.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

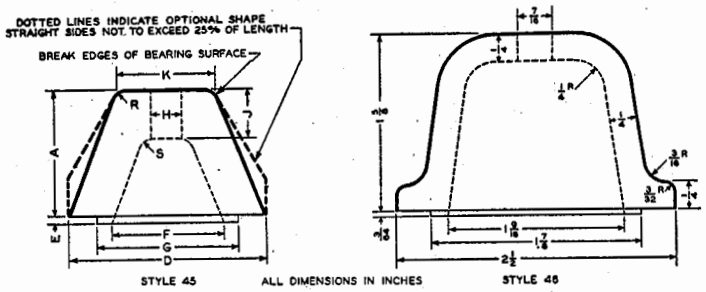
Type designation†	Dimensions (inches)							Body of insulator of grade§		
	L	D	E	F	G	H	J	L-3	L-4	L-5
								Available in finishes of—		
NS- -4001	0.425	------	0.050	------	-------	0.173	------	UW	UW	UW
NS- -4002	0.425	------	0.050	------	-------	8-32 tap	------	UW	UW	UW
NS- -4003	7/16	------	1/8	------	-------	0.200	------	UW	UW	UW
NS- -4004	1/2	------	9/16	------	-------	0.173	------	UW	UW	UW
NS- -4101	5/8	1/2	1/4	------	15/64	0.143	3/8	BW	BW	BW
NS- -4102	5/8	5/8	1/4	------	23/64	0.156	1/2	BW	BW	BW
NS- -4103	7/8	7/8	3/8	------	15/32	0.200	3/4	BW	BW	BW
NS- -4104	1 1/16	1 1/8	9/16	------	23/32	0.200	1	BW	BW	BW
NS- -4201	3/8	1/2	1/4	1/4	-------	0.143	3/8	BW	BW	BW
NS- -4202	5/8	5/8	1/4	3/8	-------	0.156	1/2	BW	BW	BW
NS- -4203	1/2	7/8	3/8	1/2	-------	0.200	3/4	BW	BW	BW
NS- -4204	3/4	1 1/8	9/16	3/4	-------	0.200	1	BW	BW	BW

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-).
§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

FIGURE 9.—Steatite radio insulators—bushing insulators.

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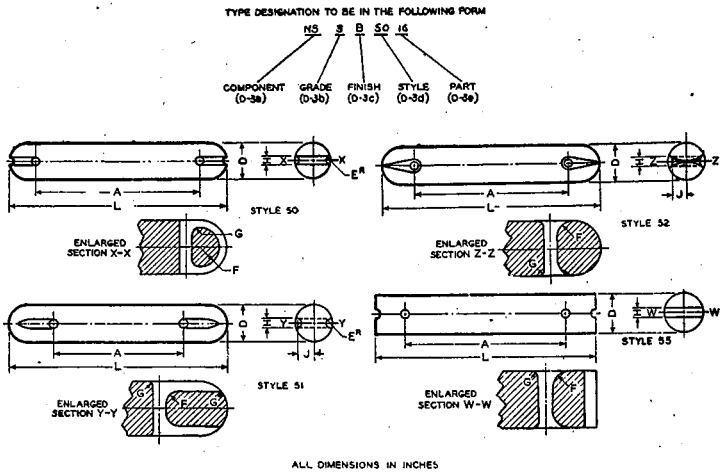
Heavy lines indicate glaze.
Glaze on inside optional with the manufacturer.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

Type designation†	Dimensions (inches)										Body of insulator of grade§		
	A	D	E	F	G	H	J	K	R	S	L-3	L-4	L-5
											Available in finishes of—		
NS--4501	7/8	1 1/4	1/32	9/16	3/4	13/64	1/4	1/2	3/32	1/8	BW	BW	BW
NS--4502	1 1/8	1 3/4	1/16	1	1 1/4	17/64	7/16	7/8	1/8	1/8	BW	BW	BW
NS--4601	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	BW	BW	BW

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-).
§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

FIGURE 10.—Steatite radio insulators—lead-in insulators.

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Heavy lines indicate glaze.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.
Samples shall be tested to determine compliance with paragraph E-7.

Type designation†	Dimensions (inches)								Minimum breaking load in tension (lbs.)	Body of insulator of grade §		
	L	A	D	E	F	G	H	J		L-3	L-4	L-5
										Available in finishes of—		
NS--5016	2¾	2	½	1/16	3/16	1/16	⅛	------	300	BW	BW	BW
NS--5124	4¼	3	½	1/16	⅛	1/16	⅛	7/32	450	BW	BW	BW
NS--5148	7¼	6	½	1/16	⅛	1/16	⅛	7/32	450	BW	BW	BW
NS--5212	3½	1½	¾	------	3/16	1/16	3/16	5/16	450	BW	BW	BW
NS--5224	4	3	¾	------	3/16	1/16	3/16	5/16	450	BW	BW	BW
NS--5240	6	5	¾	------	3/16	1/16	3/16	5/16	450	BW	BW	BW
NS--5524	4⅛	3	¾	------	3/16	1/16	¼	------	490	BW	BW	BW
NS--5540	6⅛	5	¾	------	3/16	1/16	¼	------	490	BW	BW	BW
NS--5572	10⅛	9	¾	------	3/16	1/16	¼	------	490	BW	BW	BW
NS--5596	13⅛	12	¾	------	3/16	1/16	¼	------	490	BW	BW	BW

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-).
§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

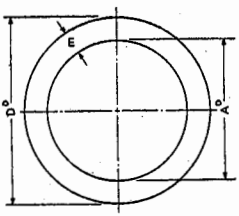
FIGURE 11.—Steatite radio insulators—spreaders and strain insulators.

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TYPE DESIGNATION TO BE IN THE FOLLOWING FORM

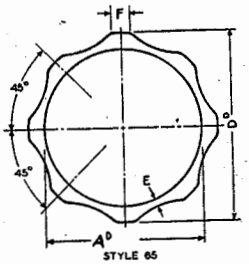
NS 3 U 60 01

COMPONENT (D-3a) GRADE (D-3b) FINISH (D-3c) STYLE (D-3d) PART (D-3e)



STYLE 60

ALL DIMENSIONS IN INCHES



STYLE 65

Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

Type designation †	Dimensions (inches)				Body of insulator of grade §		
	A	D	E (Nominal)	F (Nominal)	L-3	L-4	L-5
					Available in finishes of—		
NS--6001	5/16	1/2	5/32		UW	UW	UW
NS--6002	13/32	5/8	7/64		UW	UW	UW
NS--6003	1/2	3/4	1/8		UW	UW	UW
NS--6004	11/16	1	5/32		UW	UW	UW
NS--6005	3/4	1 1/16	3/16		UW	UW	UW
NS--6006	29/32	1 1/4	1 1/64		UW	UW	UW
NS--6007	1 1/8	1 1/2	3/16		UW	UW	UW
NS--6008	1 9/16	2	7/32		UW	UW	UW
NS--6009	2	2 1/2	1/4		UW	UW	UW
NS--6010	2 7/16	3	9/32		UW	UW	UW
NS--6011	3 5/16	4	1 1/32		UW	UW	UW
			Min. Max.	Min. Max.			
NS--6501	1 9/16	2	3/32 1/8	1/4	UW	UW	UW
NS--6502	2 1/16	2 1/2	1/8 3/16	1/4	UW	UW	UW
NS--6503	2 1/2	3	1/8 3/16	1/4	UW	UW	UW
NS--6504	3 1/4	4	5/32 7/32	3/8 1/2	UW	UW	UW

†Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-). Tubes have not been standardized with respect to number of coil-winding grooves per inch, location of lead holes, or length of coil form, but standard cross sections have been established. Under this specification, tubes are not required to carry individual part numbers but may be marked with the component and grade symbols of the type designation (D-3). Example—NS3.

§In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

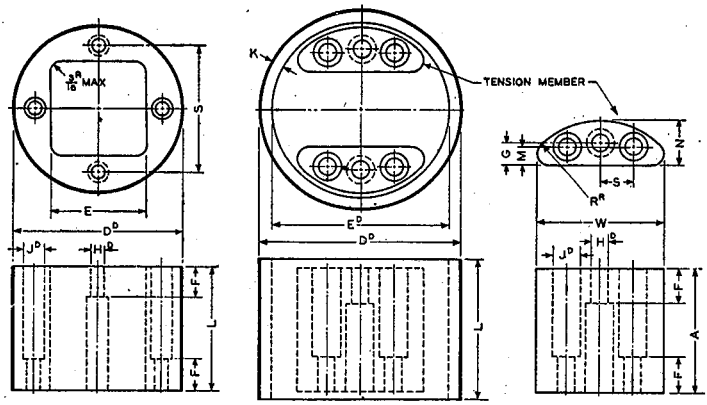
FIGURE 12.—Steatite radio insulators—tubes, coil, and miscellaneous (diameter and wall thickness).

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TYPE DESIGNATION TO BE IN THE FOLLOWING FORM

N5 3 W 75 80

COMPONENT (D-3a) GRADE (D-3b) FINISH (D-3c) STYLE (D-3d) PART (D-3e)



STYLE 70 HOUSING INSULATOR STYLE 75 STYLE 76

ALL DIMENSIONS IN INCHES

Heavy lines indicate glaze when specified.
Minimum and maximum diameters as shown include all allowances for out-of-round.
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

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Type designation †	For mica capacitor	Dimensions (inches)							Body of insulator of grade §	
									L-3	L-4
		L	D	E	F	H	J	S	Available in finishes of—	
NS- -7075	CM75	$2\begin{cases}+0.000\\-0.040\end{cases}$	$2\frac{3}{64}$ max. $2\frac{1}{2}$ min.	$1\frac{1}{2}\pm\frac{1}{32}$	$\frac{1}{2}$	$\frac{7}{32}$	$1\frac{1}{32}$	2	W	W

Type designation †	For mica capacitor	Dimensions (inches)				Body of insulator of grades §	
						L-3	L-4
		L	D	E	K min.	Available in finishes of—	
NS- -7580	CM80	$2\frac{1}{4}\begin{cases}+0.050\\-0.000\end{cases}$	$3\frac{1}{16}\pm\frac{1}{16}$	$2\frac{3}{4}\pm\frac{1}{16}$	$\frac{3}{16}$	W	W
NS- -7585	CM85	$3\frac{5}{16}\begin{cases}+0.065\\-0.000\end{cases}$	$4\frac{1}{2}\pm\frac{1}{8}$	$4\pm\frac{1}{8}$	$\frac{1}{4}$	W	W
NS- -7590	CM90	$5\frac{1}{16}\begin{cases}+0.100\\-0.000\end{cases}$	$4\frac{1}{2}\pm\frac{1}{8}$	$4\pm\frac{1}{8}$	$\frac{1}{4}$	W	W

Type designation †	For mica capacitor	Dimensions (inches)										Body of insulator of grade	
												L-3	L-4
		A	F	G	H	J	M	N	R	S	W	Available in finishes of—	
NS- -7680	CM80	$2\begin{cases}+0.000\\-0.040\end{cases}$	$\frac{9}{16}$	$\frac{3}{8}$	$1\frac{7}{64}$	$\frac{7}{16}$	$\frac{5}{16}$	$2\frac{3}{32}$	$1\frac{11}{32}$	$1\frac{7}{32}$	2	U	U
NS- -7685	CM85	$3\frac{1}{16}\begin{cases}+0.000\\-0.030\end{cases}$	$\frac{3}{4}$	$\frac{3}{8}$	$1\frac{7}{64}$	$\frac{7}{16}$	$\frac{5}{16}$	$1\frac{3}{16}$	$1\frac{15}{16}$	$\frac{3}{4}$	$2\frac{7}{8}$	U	U
NS- -7690	CM90	$4\frac{13}{16}\begin{cases}+0.000\\-0.095\end{cases}$	$\frac{3}{4}$	$\frac{3}{8}$	$1\frac{7}{64}$	$\frac{7}{16}$	$\frac{5}{16}$	$1\frac{3}{16}$	$1\frac{15}{16}$	$\frac{3}{4}$	$2\frac{7}{8}$	U	U

† Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (-).

Ceramic part of Fixed Mica-Dielectric Capacitors, Types CM80, CM85, and CM90 (JAN-C-5) consists of 3 pieces: 1 housing insulator (Style 75) and 2 tension members (Style 76).

§ In the case of glazed insulators, the grade of the combined body (basic ceramic body plus additional glaze) shall be not less than that of the unglazed body alone.

FIGURE 13.—Steatite radio insulators—Shells for types CM75 to CM90—Fixed mica dielectric capacitors.

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APPENDIX

Design Criteria

This section on design criteria was developed to be used as a reference for engineers and draftsmen in designing steatite into their equipment. The data compiled represents that which is considered good design practice and does not necessarily imply that closer tolerances on any important dimensions or special designs cannot be produced by special handling in some form or another. It is definitely recommended that the following material be carefully considered by engineers and draftsmen in designing steatite items, thus increasing production by reducing the possibility of waste, delay, and assembly difficulties. In order to insure the most desirable design and to facilitate procurement thereof at minimum cost and production time, it is recommended that designers consult the insulator manufacturer concerning specific design details applicable to the insulator under consideration.

1. *Tolerances.*—In general and unless otherwise specified, the notes below referring to tolerances apply to all dimensions of insulators. In the event that certain specific dimensions require tolerances closer than those required under "critical dimensions," they should be so indicated on each drawing.

1a. *Noncritical dimensions.*—All noncritical dimensions, which are indicated as such by common fractional dimensions on applicable drawings and orders, infer tolerances of ±2 percent of the dimension involved, but not more than ±1/8 inch, on dimensions up to and including 12½ inches. On all dimensions above 12½ inches a tolerance of ±1 percent would apply. In no case should any tolerance be less than ±1/64 inch.

1b. *Critical dimensions.*—All critical dimensions, which are indicated as such by decimal dimensions on applicable drawings or orders, infer tolerances as follows:

1b(1). *Unglazed surfaces (general).*—A tolerance of ±1 percent of the dimension involved, but not less than ±0.005 inch, applies between all unglazed surfaces. All thickness dimensions bear a minimum tolerance of ±0.010 inch, unless otherwise specified.

1b(2). *Glazed surfaces (general).*—A tolerance of ±2 percent of the dimension involved, but not less than ±0.012 inch, applies between all glazed surfaces. This tolerance applies to any dimension referenced to a glazed surface.

1b(3). *Cylindrical shapes—outside and inside diameters (unglazed).*—A tolerance of ±1 percent of the dimension involved, but not less than ±0.005 inch, applies to all unglazed diameters. All nominal wall thickness dimensions bear a minimum tolerance of ±0.010 inch, unless otherwise specified.

1b(4). *Cylindrical shapes—outside and inside diameters (glazed).*—A tolerance of ±2 percent of the dimension involved, but not less than ±0.012 inch, applies on all diameters up to and including 1 inch. Where the diameter is greater than 1 inch, ±(1 percent of the diam-

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eter involved +0.010 inch) is used. All nominal wall thickness dimensions bear a minimum tolerance of ±0.010 inch, unless otherwise specified.

Example:

Glazed tubes (inches)	*Tolerances (± inch)*
O. D. 0.500	O. D. 0.012
I. D. .125	I. D. .012
O. D. 1.125	O. D. .021
I. D. .825	I. D. .0165
O. D. 2.00	O. D. .030
I. D. 1.750	I. D. .0275

1b(5). *Camber.*—The camber shown in table I should apply to all surfaces of all shapes. Figures 1, 2, and 3 should be used to establish the ratios listed in table Ia of the appendix. For flat shapes, disks, plates, and the like, these limits should apply only where the thickness is equal to or greater than the thickness determined by the formulas given in 1b(10) of the appendix. Where the thickness is less than that determined from 1b(10) of the appendix, the camber should be specified on the applicable drawing or specification.

TABLE Ia

$\frac{L}{a}$	Maximum camber (percent of *L*) to be applied to surfaces	
	Unglazed	Glazed
Less than 2	0.25	0.35
From 2 to less than 3	.30	.40
From 3 to less than 5	.35	.45
From 5 to less than 8	.40	.50
From 8 to less than 13	.45	.55
From 13 to less than 20	.50	.60
From 20 to less than 30	.55	.65
From 30 to less than 40	.60	.70
From 40 to less than 50	.65	.75

NOTE.—In determining "maximum camber" permitted, values of e (figs. 1, 2, and 3) less than 0.006 inch should be disregarded, unless otherwise specified on the applicable figure.

1b(6). *Wall thickness of tubes.*—The wall thickness of all cylindrical shapes should be not less than 8 percent of the outside diameter nor more than 20 percent or ⅜ inch, whichever is smaller, depending upon the length of the form as compared to the outside diameter. If the length of the form is greater than 3 times the diameter, but not over 6 times the diameter, it is desirable to increase the wall thickness as much as practicable, up to 20 percent of the outside diameter. If the length is more than 6 times the outside diameter, the wall thickness should be approximately 10 percent of the outside diameter. Wall thicknesses of less than 1/32 inch or diameters smaller than 0.100 inch should be avoided.

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NOTE.—See standard O. D. and I. D. dimensions, figure 12.

1b(7). *Ellipticity of tubes.*—The ellipticity is determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient should not exceed 1.02 and applies to cylindrical shapes having a wall thickness of at least 10 percent of the outside diameter. If the wall thickness at the thinnest section of the wall is less than 10 percent of the outside diameter, the same ratio should not exceed 1.03.

1b(8). *Hole centers.*—The standard tolerance for the dimension between centers of holes should be ±0.005 inch for all hole-center spacings less than 0.500 inch, and ±1 percent of the dimension involved for all hole-center spacings of 0.500 inch and greater. If pin gages are used for inspecting hole-center spacing, the design of the gage should be such as to meet the specific requirements. In using pin gages consideration should be given to the tolerance between hole centers as well as hole-diameter tolerance.

1b(9). *Hole diameters (unglazed).*—The standard tolerance for all hole diameters should be ±0.005 inch for all diameters up to 0.500 inch, and ±1 percent of the hole diameter for all diameters 0.500 inch and greater. Holes which are not perfect circles should have the same tolerance applied to the minor axis only. Holes leading from glazed surfaces should have the tolerances for glazed surfaces applied unless the drawing requires that glaze be removed from the hole, in which case the tolerances for unglazed surfaces should apply.

1b(9)a. *Hole diameters (glazed).*—The standard tolerance for all hole diameters should be ±0.012 inch for all diameters up to 0.600 inch, and then ± 2 percent of the diameter for all diameters 0.600 inch and greater. Holes which are not perfect circles should have the same tolerance applied to the minor axis only.

1b(10). *Thickness.* ± For flat shapes, disks, plates, and the like, the minimum thickness should be computed as follows:

Where the ratio of the maximum superficial dimension to the largest dimension at right angles to it is less than 5,

$$\text{minimum thickness in inches} = \frac{\sqrt{A}}{10}$$

Where the same ratio is 5 or greater.

$$\text{minimum thickness in inches} = \frac{\sqrt{A}}{8}$$

In both cases A is the outline area.

1b(11). *Parallel limits.*—Parallelism of surfaces is considered satisfactory if the thickness measured at any point is within the established thickness tolerance of the piece.

1b(12) *Pillar insulators.*—Parallelism of top and bottom surfaces of pillar insulators is to be within 2 degrees, and these surfaces are to be perpendicular to the center axis within 2 degrees.

2. *Extruded shapes.*—

2a. Wherever the shapes required must be produced by extrusion

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it is desirable to produce as uniform a wall thickness as possible. All outside shapes should be made as symmetrical as possible and the inside preferably cylindrical.

2b. It is desirable to maintain a reasonable wall thickness wherever possible, and under no condition should walls be excessively thin or thick. Note practical limits set up under 1b(6).

2c. All holes perpendicular to the axis of extrusion are by the nature of the process somewhat elliptical. This fact should be taken into consideration by the designer.

3. *Pressed parts.*—

3a. It is recommended that all pressed parts be designed so as to maintain as nearly as possible a uniform thickness over the full area of the piece.

3b. Exceedingly small holes and blind holes should be avoided where possible.

3c. Reasonable tapers and radii (draft and fillets) should be specified for all depressions or raised bosses.

3d. Thin walls between holes and depressions or outside edges should be avoided where possible.

3e. Where parts are to be mounted on some flat surface, they should be so designed that they can be ground flat after firing to prevent breakage during the mounting operation.

3f. Elongated holes, or preferably oversize round holes, should be specified where parts are to match corresponding holes in mating parts, so as to allow for variation in dimensions of the ceramic part.

4. *Machined parts.*—Most of the points covered on pressed and extruded parts also apply to machined parts in addition, it is definitely desirable, due to the fragility of the material, to avoid all thin walls or thin barriers between recesses at the time machining operations are carried out. It is generally considered good design procedure to bear in mind that the operations involved in the machining of steatite parts are definitely comparable to those which would be required if the same parts were to be made in metal, adding, of course, the factor of fragility. In other words, machined steatite parts should be so designed that normal machine shop equipment could be used in producing the desired shape, thus avoiding any rectangular or square depressions, square holes, or the like.

5. *Tapped holes.*—It is recommended that the use of tapped holes in steatite parts be avoided if it is possible to provide other satisfactory methods of mounting. If tapped holes are specified, the following points should be taken into consideration.

5a. Due to the nature and machining characteristics of steatite material, 6–32 or larger tapped holes should be specified. The largest practicable tapped holes should be used.

5b. All tapped holes should be slightly countersunk.

5c. It is recommended that the designer, wherever possible, increase the depth of the tapped hole below that required for the screw to an amount equivalent to at least 3 threads plus the original drill tip.

5d. Effective tapped holes should be a minimum of 6 threads and a maximum of 12 threads.

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6. *Wire groove threads.*—

6a. Grooved parts used for coil supports should be measured by a three-wire method which includes two wires in adjacent grooves on one side and one wire in a groove between them on the other side of the coil form; and the specifications should include the wire diameter and the nominal diameter over wires with aforementioned diameter tolerances to apply.

6b. Where "V" threads are required, 90 degrees is considered as standard. All others are considered as being special.

6c. A tolerance of ±3 degrees should be permitted on thread angles wherever possible so as to permit normal variations in shrinkage and tool wear.

6d. A slight radius is essential at the bottom of all grooves (instead of a sharp V) so as to allow for glazing, if required, and for threading-tool wear.

7. *External screw threads for assembly purposes.*—

7a. External screw threads on steatite for assembly purposes should be avoided wherever possible.

7b. Coarse series threads should be specified where the length of thread in engagement is equal to or greater than 6 threads. Otherwise the fine series is recommended, but length of engagement in the nut less than 6 threads is not recommended. Special threads should be avoided.

8. *Miscellaneous design hints.*—

8a. *Edges.*—All sharp edges, where practicable, should include a slight radius or bevel to prevent chipping during manufacturing operations and handling.

8b. *Fastening of steatite parts.*—Where steatite parts are fastened rigidly to other parts of an assembly, it is recommended that a resilient material be provided to take up the differential expansion and thus prevent breakage of the steatite. Suspension at 3 points or less is desirable.

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